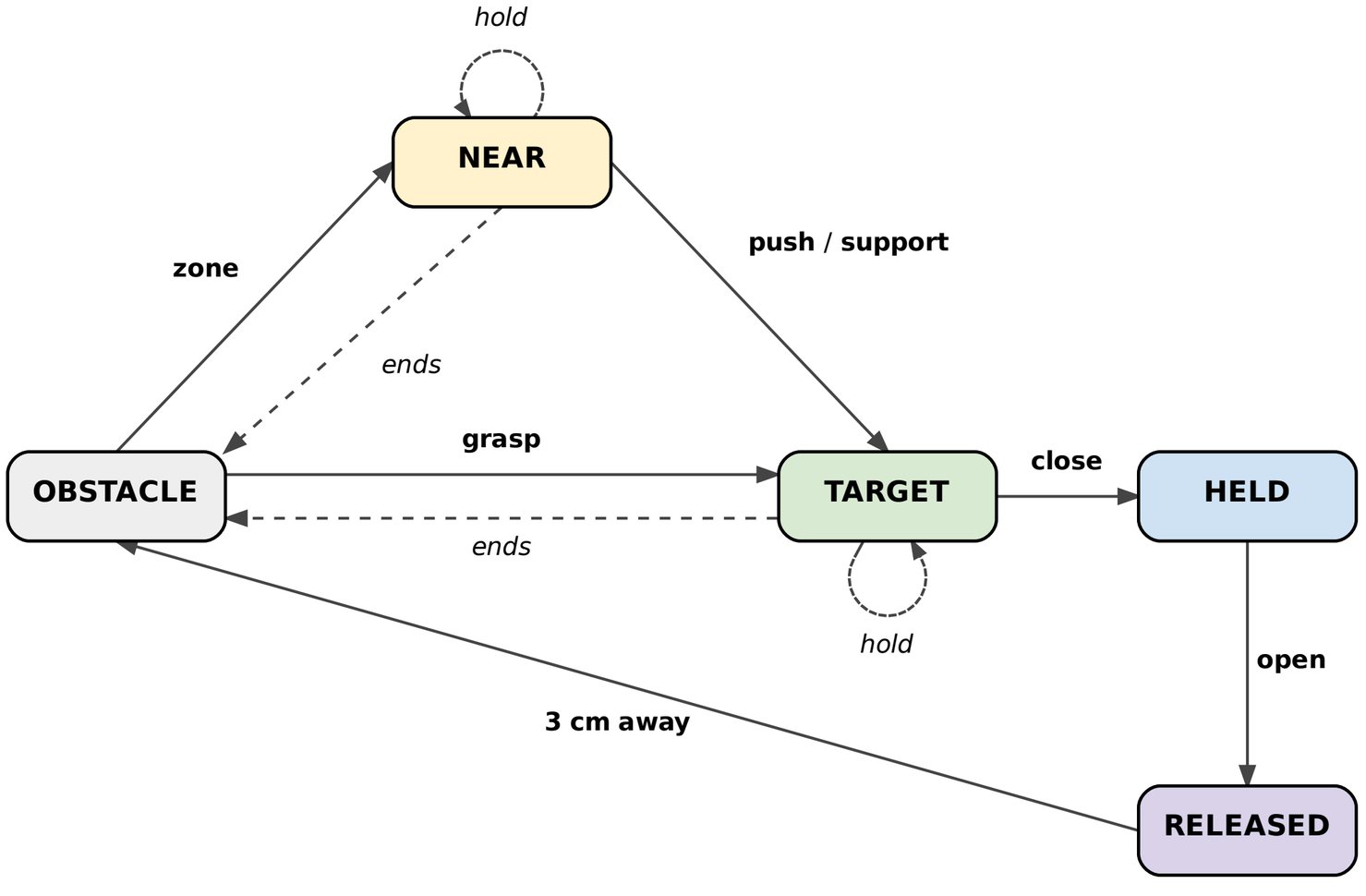
Object roles: what the robot may touch



States objects without an explicit permission; the arm always keeps its full margin	
OBSTACLE	default: full margin, halt on penetration
NEAR	hand may come 3 cm closer, halt on penetration
TARGET	hand and held object may touch it
HELD	in the gripper, moves with the hand
RELEASED	just let go, the hand may still touch it
Transitions on keyframes closing the gripper or with ignore_collisions	
zone	within 3.5 cm of the hand at the keypose
grasp	between the fingers at the keypose
push	hand empty: first object hit in 3.5 cm of travel
support	holding: brought onto it, or resting on it
close / open	the gripper closes on it / opens
3 cm away	the hand has left the released object
hold	still touched: keeps its role
ends	keyframe over and hand 3.5 cm away, or moving on its own > 3 cm/s (a pushed object may move with the hand)
Permissions config / VLM restrict, only a human relaxes	
TOUCH	human: TARGET whatever the state
AVOID	never NEAR / TARGET
FRAGILE	never NEAR / TARGET, margin +3 cm
Precedence	
HELD > RELEASED > TOUCH > FRAGILE > AVOID > TARGET / NEAR > OBSTACLE. The table is never pushed.	